

CEG 4424/6424 (Security Attacks and Defenses) Homework #2

Question 1 (10 points)

What is the difference between a block cipher and a stream cipher?

Answer: A block cipher processes plaintext in fixed-size blocks (e.g., 128-bit blocks in AES) at a time, applying the same transformation and key schedule across each entire block.

A stream cipher operates continuously on smaller units of data, typically bit-by-bit or byte-by-byte. It uses a secret key to generate a pseudorandom keystream, which is then combined with the plaintext using a bitwise XOR operation.

Question 2 (10 points)

Suppose that you are given a plaintext message (P) and its corresponding ciphertext C using One-Time Pad with a specific key (K), i.e., $C = K \text{ XOR } P$, can you compute the One-Time Pad key (K) from P and C ? (Hint: using P and C , to compute K)

Question 2:- Computing the One-Time Pad Key
Ans:- Yes, you can directly compute the One-Time Pad key (K) from the plaintext (P) and ciphertext (C)
Because the bitwise XOR operation is its own inverse ($A \oplus B = C \Rightarrow A \oplus C = B$),
XORing the ciphertext with the plaintext recovers the key: $K = C \oplus P$

Answer

Question 3 (10 points)

Suppose we have a computer that can do 2^{50} AES decryptions per second. What is the expected time (in seconds) to find the correct key by brute force search if the AES key setting is 128 bits?

Q no 3:- AES Brute Force Search Time

- Total key space for AES-128: 2^{128} possible keys.
- Expected (average) search space: $\frac{2^{128}}{2} = 2^{127}$ keys.
- Decryption rate: 2^{50} decryption per second.

$$\text{Expected Time} = \frac{2^{127} \text{ keys}}{2^{50} \text{ keys/second}} = 2^{77} \text{ seconds}$$

In decimals:

$$2^{77} \approx 1.511 \times 10^{23} \text{ seconds.}$$

ANSWER:

Question 4 (10 points). Show the first eight words of the key expansion for a 128-bit key
0x 0F 0E 0D 0C 0B 0A 09 08 07 06 05 04 03 02 01 00.

Answer:

Question 4:- Key Expansion

Given:
128-bit key hex: 0F 0E 0D 0C 0B 0A 09 08 07 06 05 04
03 02 01 00.

The initial key is split into four 32-bit (4 byte) words
(w_0 through w_3):

- $w_0 = 0x0F0E0D0C$
- $w_1 = 0x0B0A0908$
- $w_2 = 0x07060504$
- $w_3 = 0x03020100$

To calculate w_4 :-

1. $w_3 = 0x03020100$

2. RotWord(w_3): 02 03 00 03

3. SubWord using AES S-box

- S-box(02) = 77

- S-box(03) = 7C

- S-box(00) = 63

- S-box(03) = 78

Result = 777C6378

4. XOR with round constant [1] = $0x02000000$:

$$0x777C6378 \oplus 0x02000000$$

$$= 0x767C6378$$

5. XOR with w_0 :

$$w_4 = 0x0F0E0D0C \oplus 0x767C6378$$

$$= 0x79726E77$$

Next words ($w_i = w_{i-2} \oplus w_{i-4}$):

- $w_5 = w_4 \oplus w_1 = 0x79726E77 \oplus 0x0B0A0908 = 0x7278677F$

- $w_6 = w_5 \oplus w_2 = 0x7278677F \oplus 0x07060504 = 0x757E627B$

- $w_7 = w_6 \oplus w_3 = 0x757E627B \oplus 0x03020100 = 0x767C6378$

First 8 words:-

$w_0 = 0x0F0E0D0C$

$w_1 = 0x0B0A0908$

~~$w_2 = 0x07060504$~~

$w_3 = 0x03020100$

$w_4 = 0x79726E77$

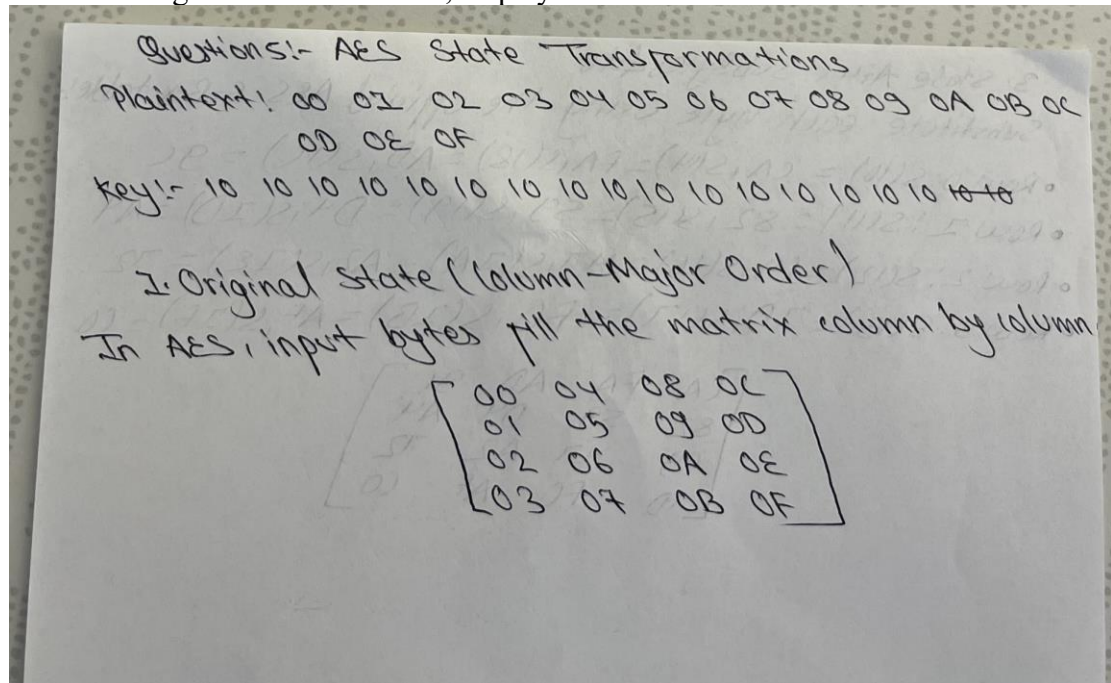
$w_5 = 0x7278677F$

$w_6 = 0x757E627B$

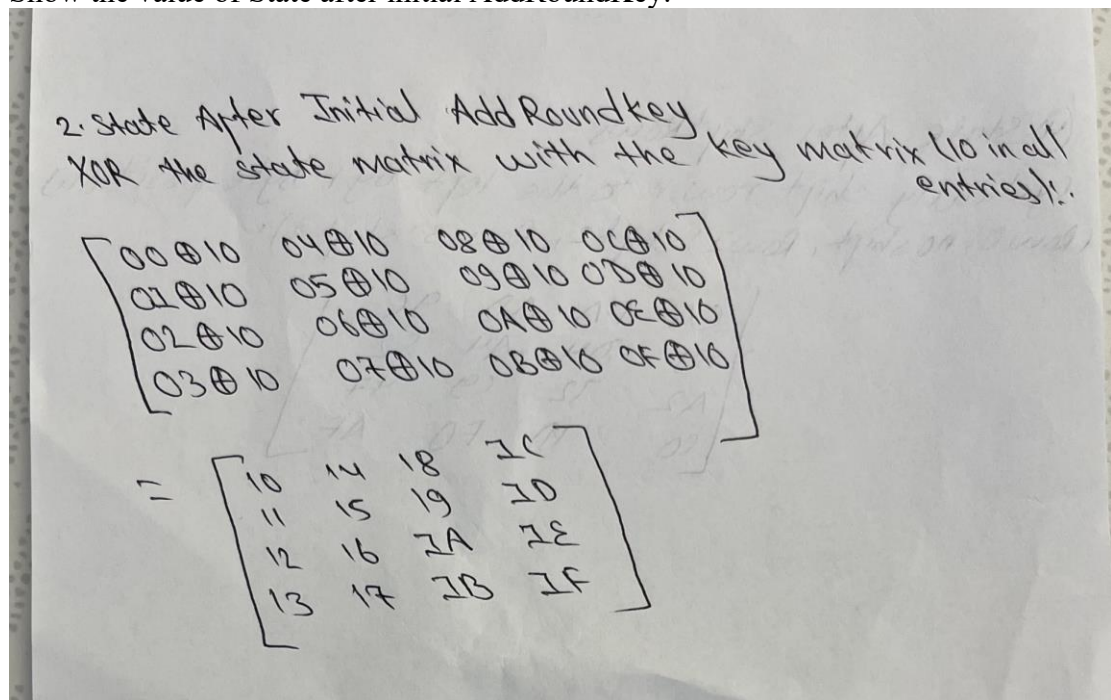
$w_7 = 0x767C637B$

Question 5 (20 points) Given the plaintext {000102030405060708090A0B0C0D0E0F} and the key {10101010101010101010101010101010},

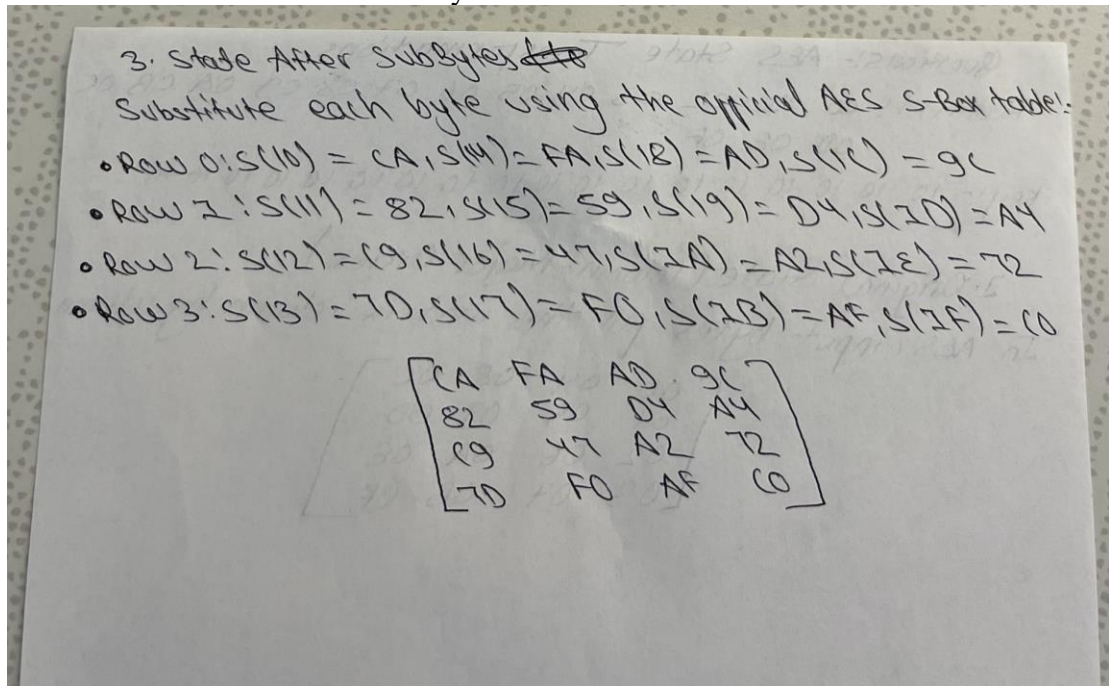
(1) Show the original content of State, displayed as a 4*4 matrix.



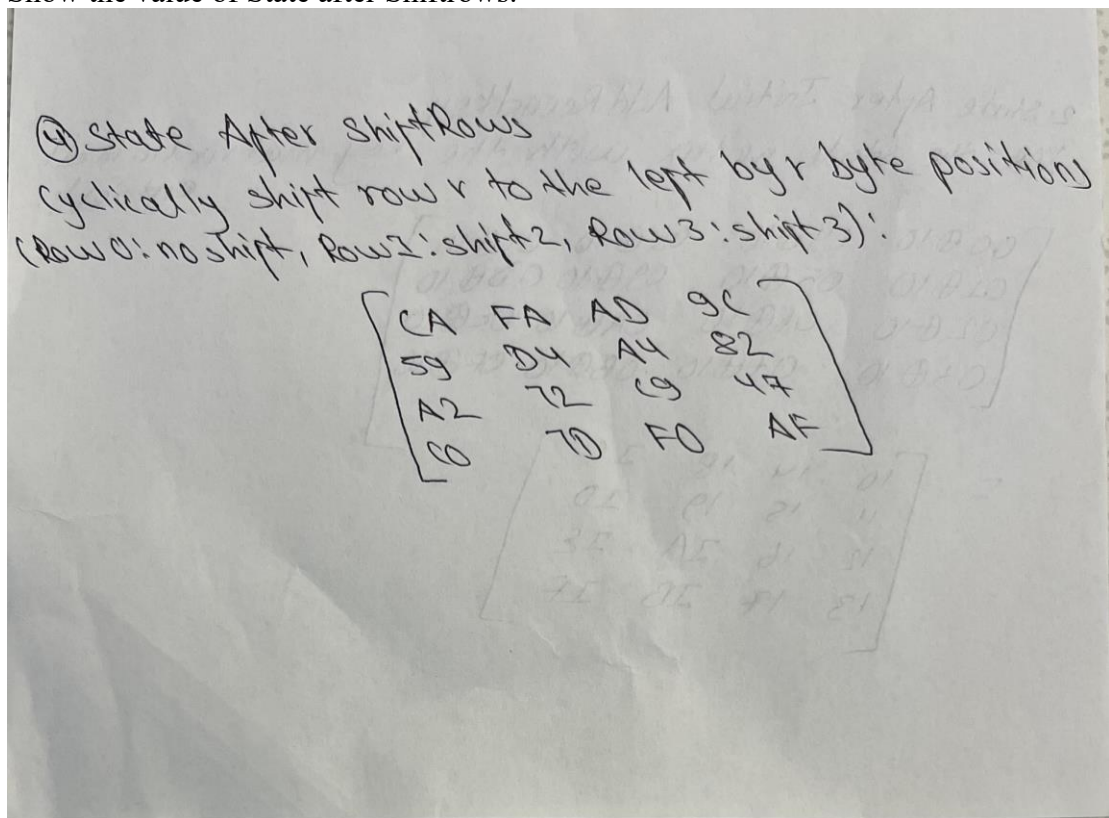
(2) Show the value of State after initial AddRoundKey.



(3) Show the value of State after Subbytes.



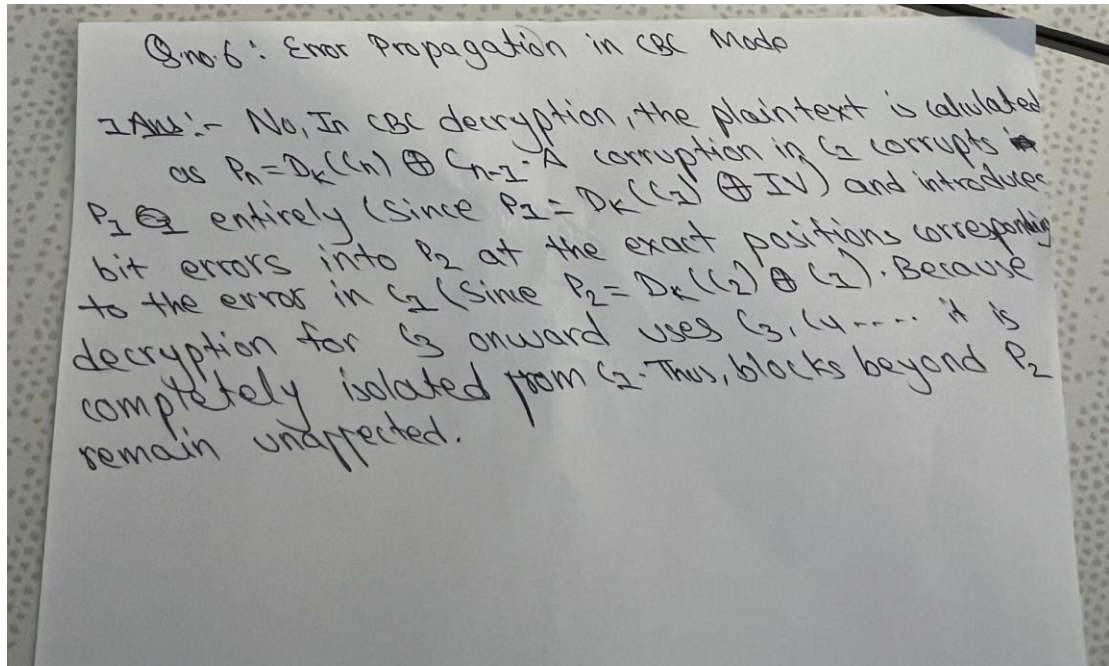
(4) Show the value of State after Shiftrows.



Question 6 (10 points). With the ECB mode of DES, if there is an error in a block of the transmitted ciphertext, only the corresponding plaintext block is affected. However, in the

CBC mode, this error propagates. For example, an error in the transmitted C_1 obviously corrupts P_1 and P_2 .

- (a) Are any blocks beyond P_2 affected if there is an error in the transmitted C_1 in the CBC mode?



- (b) Suppose that there is a bit error in the source version of P_1 . Through how many ciphertext blocks is this error propagated in the CBC mode?

2 Ans:- All subsequent ciphertext blocks (C_2 through C_n). In CBC encryption, $C_1 = E_K(P_1 \oplus IV)$. Because each ciphertext block feeds directly into the next encryption stage ($C_2 = E_K(P_2 \oplus C_1)$, $C_3 = E_K(P_3 \oplus C_2)$, etc.) any error in P_2 changes C_2 , which cascades forward into every subsequent block.

Question 7 (10 points). The Common Weakness Enumeration (CWE) system provides a reference method for publicly known information-security vulnerabilities and exposures. Please describe what is CWE-1240, CWE-311. (Hint: Please go to the website cve.mitre.org to find the descriptions of these CWEs. You should describe the content in your own words by rephrasing the sentences from the online resources.)

Answer: Common Weakness Enumeration (CWE) Descriptions

- **CWE-1240 (Use of a Cryptographic Primitive with a Risky Implementation):** This vulnerability occurs when a software or hardware design relies on a custom, unverified, or non-compliant implementation of a cryptographic primitive instead of using established, standard implementations. Proprietary or poorly reviewed cryptographic code frequently introduces subtle operational or mathematical flaws that weaken the overall security of the system.
- **CWE-311 (Missing Encryption of Sensitive Data):** This weakness happens when an application stores or transmits sensitive details (such as login credentials, personally identifiable information, or financial records) in plaintext without proper encryption. Without protective encryption in place, adversaries can easily read or intercept the sensitive data.